Bodie W. Cooper
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Education:

- James Madison University, Harrisonburg, VA
- Bachelor of Integrated Science and Technology
- Tailored Concentration Applied Computing & Industrial Manufacturing Systems

Technical Skills:

Coding/Programs Used:

- Python for various tasks
- SQL & HTML for Capstone Project
- Worked with JavaScript
- CAD in SolidWorks & Fusion360
- RStudio Statistical Environment
- Jupyter Notebook
- Google CoLab

Operating Systems:

- Windows
- Raspberry Pi

Machinery:

- Laser Engraver
- CNC Turner
- Circuit Cutter

Projects:

Digital Automation Capstone Project: For our senior capstone project, my partner and I have partnered with SafeRides, a student-run transportation service, to design and implement a more efficient and automated system for their operations. The purpose of this project is to streamline scheduling, ride requests, and communication to improve both rider convenience and driver coordination. To achieve this, we are applying a range of technological and analytical skills, which include the following.

a.) *Google Form Systems:* To collect and analyze student feedback on the current service and proposed improvements.
b.) *Database management:* for organizing ride data, scheduling logistics, and storing user information securely.
c.) *SQL & HTML Coding:* Build and connect backend functionality with a user-friendly web interface for ride requests and administrative oversight.

SolidWorks CAD Design: This was a design product that I created freshman year. The task was to reverse-engineer a consumer product by using precision measurement techniques and digital modeling. The assignment required measuring the physical dimensions of the object, creating an accurate engineering drawing, and then modeling it inside SolidWorks. The item I selected for this was a pair of Apple AirPods.

a.) Measurement Tools: Used calipers and precision instruments to obtain accurate measurements for the product dimensions. Detailed technical schematics were produced based on collected data and measurements.
b.) 3D Modeling: Designed and rendered a fully realized digital model of AirPods using SolidWorks CAD software.

Laser Engraving Product Design: I completed a hands-on design project through JMU's Makery that focused on exploring advanced manufacturing equipment, then creating a product that could be marketed for general consumption. This project required selecting a machine, learning its operation, and producing a personalized product. For this project, I designed a fully functional clock featuring a custom laser-engraved image of my family.

a.) Manufacturing Equipment: Operated a professional-grade laser engraving machine within JMU's Makery.
b.) Product Design: Designed and created a functional, marketable consumer product.
c.) *Personalization & Creativity*: Integrated a personalized engraved image into the final product for unique customer appeal.

Relevant Coursework:

- Programming Fundamentals
- Information Security & Privacy
- Operating Systems
- Machine Learning
- Data Analysis
- Biotechnology Issues
- Calculus with Function 1/2/3
- Cyber-Physical Industry
- Emerging Topics with Data Science
- Engineering Decisions/Engineering Opportunities
- Foundation of Lean and Six Sigma
- Database Design, Implementation & Management
- AR/VR 360 Media
- Sustainable Manufacturing Systems
- Industrial Environmental Management
- Material and Manufacturing Processes
- Programming and Problem Solving
- Topics in Applied Physics
- Topics in Applied Statistics

Working Experience:

a. Gino's Pizza, Dishwasher/Server: (2016-2025)

- Began as a dishwasher in 2016 and was promoted to server after 2 years after demonstrating strong customer service, reliability, and teamwork.
- Provided high-quality table service in a fast-paced environment, balancing accuracy, efficiency, and hospitality.
- Build rapport with diverse guests, anticipate needs, and resolve issues quickly to ensure a positive dining experience.
- Manage multiple tables and prioritize tasks effectively, consistently maintaining order accuracy and guest satisfaction.
- Process payments and handle cash/POS transactions.
- Train and support new staff, fostering a collaborative and professional team environment.
- Continue serving in a part-time capacity while attending school, demonstrating time management and commitment to both work and academics.

Relevant Skills:

- Product Design
- Manufacturing Process Improvement
- Prototyping
- Computer-Aided Design
- Technical Drawings
- Information Technology
- Databasing
- Python, HTML & SQL proficiencies
- Systems Design
- Customer Service/Hospitality Skills
- Interpersonal Communication
- Presentation Development
- Customer Retention
- Public Speaking
- Problem Solving
- Communication